

## DBMS LAB 2

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**Aim:** To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

### 1. Identify all entity types and entity sets.

**Researcher-** Represents all researchers working in the university.

**Department-** Represents all departments in the university.

**Research Project-** Represents all research projects undertaken by the university.

**Student-** Represents all students working under researchers.

**Publication-** Represents all publications produced from research projects.

**Project Assistant-** Represents all assistants associated with research projects.

### 2. Identify strong and weak entity sets.

Strong and Weak Entity Sets

- Strong Entity Sets:
  - Researcher
  - Department
  - Research Project
  - Student
  - Publication
- Weak Entity Set:
  - Project Assistant

**Project Assistant** is a weak entity because it is identified only within the context of a particular Research Project.

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### 3. Identify the owner entity and partial key of the weak entity.

**Weak Entity:** Project Assistant

**Owner Entity:** Research Project

**Partial Key:** Assistant\_Name

The complete identification of a Project Assistant is:

**Project\_ID + Assistant\_Name**

### 4. Simple attributes.

The simple attributes in the University Research Management System are:

- **Researcher:** Researcher\_ID, Name, Date\_of\_Birth, Email
- **Department:** Department\_ID, Department\_Name, Location
- **Research Project:** Project\_ID, Project\_Title, Start\_Date, End\_Date, Budget
- **Student:** Student\_ID, Name, Email, Date\_of\_Birth
- **Publication:** Publication\_ID, Title, Publication\_Date, DOI, Publisher
- **Project Assistant:** Assistant\_Name, Role, Joining\_Date

### 5. Identify composite attributes.

The composite attribute in the system is:

#### 1. Researcher → Address

- Street
- City
- State
- PIN\_Code

**Address** is a composite attribute because it can be divided into smaller component attributes.

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**Name** is treated as a simple attribute because the given problem statement does not specify First\_Name, Middle\_Name, and Last\_Name.

### **6. Identify single-valued attributes.**

#### **Researcher**

- Researcher\_ID
- Name
- Date\_of\_Birth
- Email
- Address

#### **Department**

- Department\_ID
- Department\_Name
- Location

#### **Research Project**

- Project\_ID
- Project\_Title
- Start\_Date
- End\_Date
- Budget

#### **Student**

- Student\_ID
- Name
- Email
- Date\_of\_Birth

#### **Publication**

- Publication\_ID
- Title
- Publication\_Date
- DOI
- Publisher

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### Project Assistant

- Assistant\_Name
- Role
- Joining\_Date

### 7. Identify multi-valued attributes.

The multi-valued attributes in the system are:

#### 1. Researcher → Phone\_Number

A researcher can have multiple phone numbers.

#### 2. Research Project → Keyword

A research project can have multiple keywords.

### 8. Identify stored attributes.4

#### Researcher

- Researcher\_ID
- Name
- Date\_of\_Birth
- Email
- Phone\_Number
- Address
  - Street
  - City
  - State
  - PIN\_Code

#### Department

- Department\_ID
- Department\_Name
- Location

### Research Project

- Project\_ID
- Project\_Title
- Start\_Date
- End\_Date
- Budget
- Keyword

### Student

- Student\_ID
- Name
- Email
- Date\_of\_Birth

### Publication

- Publication\_ID
- Title
- Publication\_Date
- DOI
- Publisher

### Project Assistant

- Assistant\_Name
- Role
- Joining\_Date

## 9. Identify derived attributes.

### Derived Attribute – Age

**Age** is derived from **Date\_of\_Birth**.

Therefore, Age does not need to be stored separately in the database.

In the ER diagram, **Age** is represented using a **dashed oval**.

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### 10. Identify key attributes.

| Entity            | Key Attribute               | Key Status    |
|-------------------|-----------------------------|---------------|
| Researcher        | Researcher_ID               | Primary Key   |
| Department        | Department_ID               | Primary Key   |
| Research Project  | Project_ID                  | Primary Key   |
| Student           | Student_ID                  | Primary Key   |
| Publication       | Publication_ID              | Primary Key   |
| Project Assistant | Assistant_Name              | Partial Key   |
| Project Assistant | Project_ID + Assistant_Name | Composite Key |

### 11. Identify candidate keys and select appropriate primary keys.

| Entity            | Candidate Key               | Selected Primary Key        |
|-------------------|-----------------------------|-----------------------------|
| Researcher        | Researcher_ID               | Researcher_ID               |
| Department        | Department_ID               | Department_ID               |
| Research Project  | Project_ID                  | Project_ID                  |
| Student           | Student_ID                  | Student_ID                  |
| Publication       | Publication_ID              | Publication_ID              |
| Project Assistant | Project_ID + Assistant_Name | Project_ID + Assistant_Name |

### 12. Identify at least one composite key.

The composite key is:

**Project Assistant → Project\_ID + Assistant\_Name**

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**Assistant\_Name** alone cannot uniquely identify an assistant because the same assistant name may occur in different projects.

Therefore:

**Composite Key = (Project\_ID, Assistant\_Name)**

Another composite key occurs in the relational table created for the M:N **WORKS\_ON** relationship:

**(Researcher\_ID, Project\_ID)**

**13. Identify appropriate foreign keys after converting the ER model into relations.**

| Relation                 | Foreign Key   | References                   |
|--------------------------|---------------|------------------------------|
| <b>RESEARCHER</b>        | Department_ID | DEPARTMENT(Department_ID)    |
| <b>WORKS_ON</b>          | Researcher_ID | RESEARCHER(Researcher_ID)    |
| <b>WORKS_ON</b>          | Project_ID    | RESEARCH_PROJECT(Project_ID) |
| <b>STUDENT</b>           | Researcher_ID | RESEARCHER(Researcher_ID)    |
| <b>PUBLICATION</b>       | Project_ID    | RESEARCH_PROJECT(Project_ID) |
| <b>PROJECT_ASSISTANT</b> | Project_ID    | RESEARCH_PROJECT(Project_ID) |
| <b>RESEARCHER_PHONE</b>  | Researcher_ID | RESEARCHER(Researcher_ID)    |
| <b>PROJECT_KEYWORD</b>   | Project_ID    | RESEARCH_PROJECT(Project_ID) |

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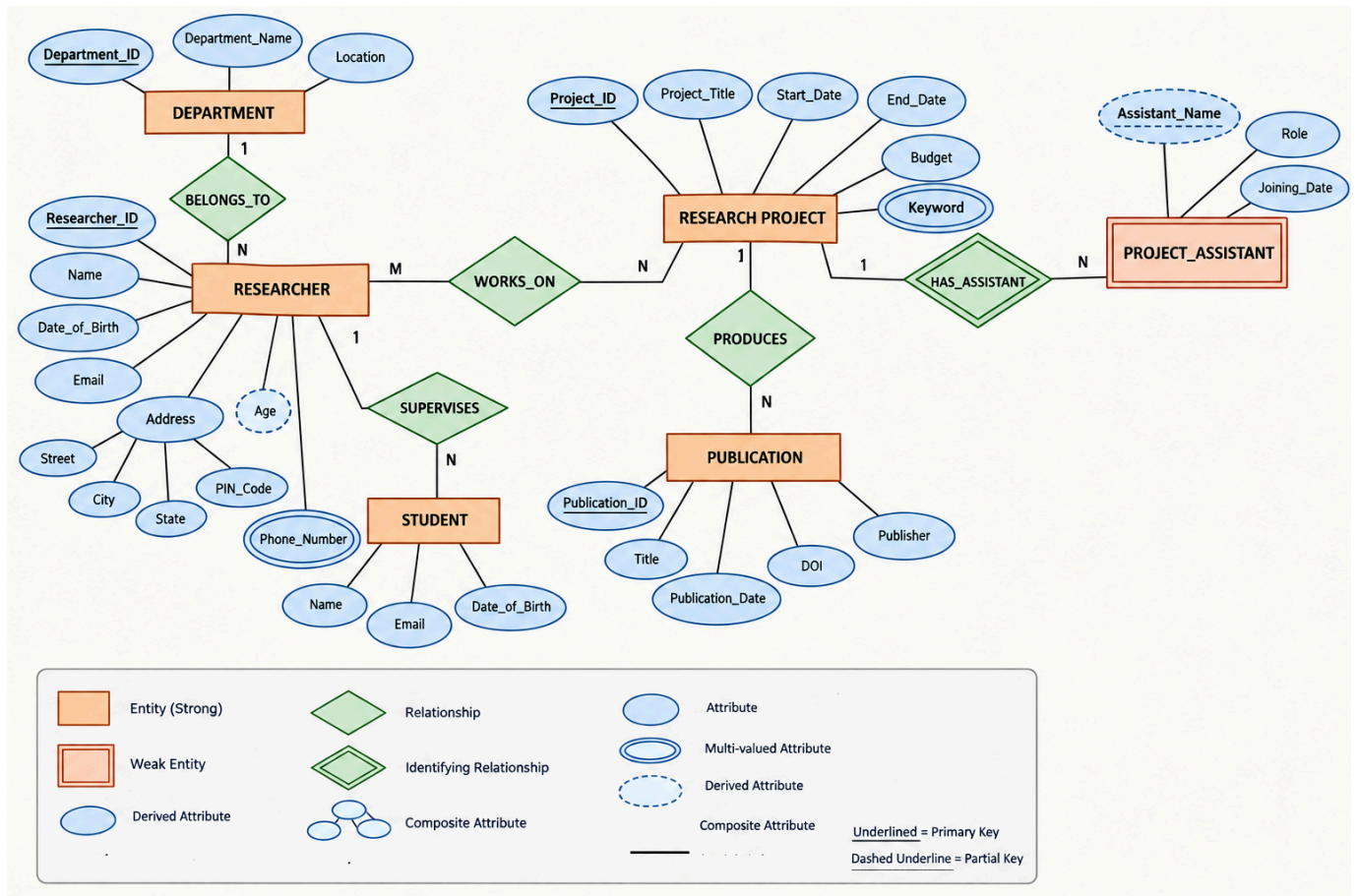
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14. Identify all major relationships between the entities.

| Relationship         | Entities Involved                    | Cardinality | Meaning                                                                                                                                                      |
|----------------------|--------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BELONGS_TO</b>    | Department ↔ Researcher              | 1 : N       | One department has many researchers; each researcher belongs to one department.                                                                              |
| <b>WORKS_ON</b>      | Researcher ↔ Research Project        | M : N       | A researcher can work on many projects, and a project can involve many researchers.                                                                          |
| <b>SUPERVISES</b>    | Researcher ↔ Student                 | 1 : N       | One researcher can supervise many students; each student works under one researcher.                                                                         |
| <b>PRODUCES</b>      | Research Project ↔ Publication       | 1 : N       | One project can produce many publications; each publication belongs to one project.                                                                          |
| <b>HAS_ASSISTANT</b> | Research Project ↔ Project Assistant | 1 : N       | One project can have many assistants; each assistant belongs to one project. This is an identifying relationship because Project Assistant is a weak entity. |



## 15. Draw the complete ER diagram using standard ER notation.



## 16. Prepare a table showing each attribute and its attribute type and key status.

| Entity     | Attribute     | Attribute Type | Value        | Key Status  |
|------------|---------------|----------------|--------------|-------------|
| Researcher | Researcher_ID | Simple         | Single       | Primary Key |
| Researcher | Name          | Simple         | Single       | Non-key     |
| Researcher | Date_of_Birth | Simple         | Single       | Stored      |
| Researcher | Age           | Derived        | Single       | Derived     |
| Researcher | Email         | Simple         | Single       | Non-key     |
| Researcher | Phone_Number  | Simple         | Multi-valued | Non-key     |
| Researcher | Address       | Composite      | Single       | Non-key     |
| Researcher | Street        | Simple         | Single       | Non-key     |

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| Entity                  | Attribute       | Attribute Type   | Value        | Key Status         |
|-------------------------|-----------------|------------------|--------------|--------------------|
|                         |                 | component        |              |                    |
| <b>Researcher</b>       | City            | Simple component | Single       | Non-key            |
| <b>Researcher</b>       | State           | Simple component | Single       | Non-key            |
| <b>Researcher</b>       | PIN_Code        | Simple component | Single       | Non-key            |
| <b>Department</b>       | Department_ID   | Simple           | Single       | <b>Primary Key</b> |
| <b>Department</b>       | Department_Name | Simple           | Single       | Non-key            |
| <b>Department</b>       | Location        | Simple           | Single       | Non-key            |
| <b>Research Project</b> | Project_ID      | Simple           | Single       | <b>Primary Key</b> |
| <b>Research Project</b> | Project_Title   | Simple           | Single       | Non-key            |
| <b>Research Project</b> | Start_Date      | Simple           | Single       | Stored             |
| <b>Research Project</b> | End_Date        | Simple           | Single       | Stored             |
| <b>Research Project</b> | Budget          | Simple           | Single       | Stored             |
| <b>Research Project</b> | Keyword         | Simple           | Multi-valued | Non-key            |
| <b>Student</b>          | Student_ID      | Simple           | Single       | <b>Primary Key</b> |
| <b>Student</b>          | Name            | Simple           | Single       | Non-key            |
| <b>Student</b>          | Email           | Simple           | Single       | Non-key            |
| <b>Student</b>          | Date_of_Birth   | Simple           | Single       | Stored             |
| <b>Publication</b>      | Publication_ID  | Simple           | Single       | <b>Primary Key</b> |

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| Entity            | Attribute        | Attribute Type | Value  | Key Status  |
|-------------------|------------------|----------------|--------|-------------|
| Publication       | Title            | Simple         | Single | Non-key     |
| Publication       | Publication_Date | Simple         | Single | Stored      |
| Publication       | DOI              | Simple         | Single | Non-key     |
| Publication       | Publisher        | Simple         | Single | Non-key     |
| Project Assistant | Assistant_Name   | Simple         | Single | Partial Key |
| Project Assistant | Role             | Simple         | Single | Non-key     |
| Project Assistant | Joining_Date     | Simple         | Single | Non-key     |